Tranexamic acid dosing strategies and blood loss reduction in multilevel spine surgery: a systematic review and network meta-analysis

Supplementary File

Supplemental Table I. Risk of Bias for Retrospective Cohort Studies

		Perez-Roman, 2019	Xue, 2018	Yu, 2017	Pong, 2018	Todeschini, 2020	Pernik, 2020
Research Question	The study addresses an appropriate and clear question	W	W	W	W	W	W
Selection of Subjects	Source populations are comparable in all respects other than the factor under investigation	W	W	W	AA	AA	AA
	Participants and those lost to follow up are compared by exposure status	N	N	N	N	NA	N
	Clearly defined outcomes	W	W	W	W	W	AA
	Recognition that exposure status could have influenced the assessment of outcome	N	N	N	N	N	N
	Reliable method of exposure	W	W	W	W	W	W
	Method of outcome assessment is valid and reliable	N	N	N	AA	N	W
Confounding	Confounders are taken into account in design and analysis	W	PA	PA	AA	AA	PA
Overall Assessment of the Study	How well was the study done to minimize bias?	Acceptable (+)	Acceptable (+)	Acceptable (+)	Acceptable (+)	Acceptable (+)	Acceptable (+)
	If coded as +, or - what is the likely direction in which bias might affect the study results?	FTT	FTT	Neither	FTT	Neither	FTT
	Certainty that overall effect is due to the study intervention?	No	No	Yes	No	Yes	Yes
	Are the results of this study directly applicable to the patient group	DA	DA	DA	DA	DA	DA

	targeted by this guideline?						
	Summary of the author's conclusions	TXA reduces operation time and postoperative blood loss and is safe at 30 days in patients undergoing PCDF.	TXA can reduce the need for blood transfusion as well as reduce postoperative blood loss and hospital stay.	TXA was associated with significantly reduced intra- and post-operative blood loss	Tranexamic acid decreased total blood loss volume and transfusion units in adult spinal deformity surgery patients	IV TXA in multilevel spine surgery may reduce intraoperative blood loss and transfusion	TXA can be helpful in reducing intra- and post-operative blood loss and transfusion for patients undergoing complex thoracolumbar deformity surgery
	Level of Evidence	2+	2+	2+	2+	2+	

Well Covered (W), Adequately Addressed (AA), Poorly Addressed (PA), Not covered (N), Directly Applicable (DA), Favors Test Treatment (FTT), Acceptable-quality cohort study (2+)

Supplemental Table II. Risk of Bias for Randomized Controlled Trials

		Peters, 2015	Mu, 2019	Shakeri, 2018	Carabini, 2018	Colomina, 2017	Farrokhi, 2011	Tsutsumimoto 2011
Research Question	The study addresses an appropriate and clear question	W	W	W	W	W	W	W
Selection of Subjects	Randomized assignment	W	W	W	W	W	W	AA
	Adequate concealment method used	AA	N	W	AA	W	W	N
	Double Blind	AA	N	W	AA	W	W	
	Groups are similar at start of trial	PA	W	W	AA	AA	W	W
	Only difference between groups is treatment under investigation	AA	W	W	W	W	W	W
	Relevant outcomes measured in a standard, valid and reliable way	PA	W	PA	PA	W	W	W
	% of individuals lost at follow up	0% TXA 7% Control	4% TXA 2% Control	0% TXA 0% Control	0% TXA 0% Control	0% TXA 2% Control	0% TXA 0% Control	0% TXA 0% Control
	Intention to treat analysis	AA	AA	W	W	W	AA	AA
	Results are comparable at all sites	NA	NA	NA	NA	Not reported	NA	NA
Overall Assessment of the Study	How well was the study done to minimize bias?	Acceptable (+)	Acceptable (+)	Acceptable (+)	Acceptable (+)	Acceptable (+)	High Quality (++)	Acceptable (+)
	If coded as +, or - what is the likely direction in which bias might affect the study results?	FTT	Neither	Neither	FTT	Neither	Neither	Neither
	Certainty that overall effect is due to the study intervention?	No	Yes	Yes	No	Yes	Yes	No
	Are the results of this study directly applicable to the patient group targeted by this guideline?	DA	DA	DA	DA	DA	DA	DA

	Summary of the author's conclusions	A trend was found in reducing intraoperative blood loss in antifibrinolytic arms when compared to control. Major weakness for including in the quantitative meta-analysis is that measures of spread (standard deviation/IQR) are not included for any outcomes. There were important differences between groups at baseline and the sample was not appropriately powered	TXA reduces blood loss in double-segment PLIF surgeries. Major weakness is that blinding/concealment was not performed in this study; otherwise, bias seems to be addressed relatively well	Intraoperative and postoperative blood loss is decreased with TXA usage. Major weakness is that the methods for calculating blood loss are not described; however, this was rigorously blinded so it's unlikely that this systematically favored either group	Total RBC transfused was decreased in TXA patient group. Major weaknesses of this study are that the methods for calculating blood loss are not described, and the placebo group had a significantly higher risk of major transfusion at baseline; the study generally controls for bias well, but has some weaknesses which may impact the observed effects	No significant effects were noted in the TXA patient group. No major weaknesses for bias identified	No significant effects were noted in the TXA patient group. No major weaknesses for bias identified	TXA reduced postoperative blood loss in cervical laminoplasty. Major weakness is that the study was not blinded; however, detailed/objective methods for calculating blood loss outcomes were provided, so bias appears to be controlled for relatively well
	Level of Evidence	1+	1+	1+	1+	1+	1++	1+

Well Covered (W), Adequately Addressed (AA), Poorly Addressed (PA), Not covered (N), Directly Applicable (DA), Favors Test Treatment (FTT),

Acceptable-quality controlled trial (1+), High-quality controlled trial (1++)